

Lab 7 – Transient Response

Abdon Morales

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Description

In this lab we study the transient response of a first-order RC circuit by using it as the timing element of an op-amp relaxation oscillator. The RC time constant sets the rate at which the capacitor voltage charges and discharges, and the op-amp, operating as a comparator in saturation, flips its output each time the capacitor voltage crosses the reference set by a resistive feedback divider. The output is therefore a square wave whose period depends only on the RC time constant and the divider ratio. We use this oscillator to alternately flash a red and a green LED at a chosen frequency. In Task 1 we build the oscillator with fixed resistors sized for a 1 Hz blink rate and measure the two input node voltages on the scope. In Task 2 we replace the divider with a 10 k Ω potentiometer and tune the circuit to 2 Hz, first at the original RC value and then with the feedback resistor halved.

Procedure

1. Measured each resistor and capacitor with the bench DMM/LCR before building, to record actual component values and rule out a bad part as a later source of error.
2. Built the relaxation oscillator from the prelab schematic on the breadboard: a 741 op-amp powered from 5 V and -5 V rails, the feedback divider R_1 (7.5 k Ω) and R_2 (2.4 k Ω) from the output to the non-inverting input, and the RC branch ($R = 10$ k Ω , $C = 100$ μ F) from the output to the inverting input.
3. Connected the red and green LEDs in anti-parallel across the output through a current-limiting resistor, so one LED conducts on the positive output swing and the other on the negative swing.
4. Powered the circuit and confirmed the two LEDs alternated at roughly one blink per second.
5. **Task 1.** Probed the non-inverting node $V_1(t)$ and the inverting node $V_2(t)$ with the oscilloscope and recorded, for each: the oscillation frequency and the peak-to-peak voltage. Captured both waveforms.
6. **Task 2, Step 1.** Replaced R_1 and R_2 with the 10 k Ω potentiometer wired as a divider, then adjusted the wiper until the circuit oscillated at 2 Hz. Captured the waveform, removed the pot, and measured the two resistances R_1 and R_2 that the wiper produced. Computed the corresponding $\gamma = R_2/(R_1 + R_2)$.
7. **Task 2, Step 2.** Replaced the RC feedback resistor R with half its value (5 k Ω), reinserted the potentiometer, and re-adjusted the wiper for 2 Hz. Captured the waveform, removed and measured R_1 and R_2 , and computed the new γ .

Calculations

Circuit operation. Because the RC feedback makes V_1 and V_2 depend on the output in different ways, the two inputs are never equal and the op-amp runs in saturation, giving an output $V_o = \pm V_{sat}$. The non-inverting input is set by the resistive divider, with no current into the op-amp:

$$\frac{V_1 - 0}{R_2} + \frac{V_1 - V_o}{R_1} = 0 \Rightarrow V_1 = V_o \frac{R_2}{R_1 + R_2} = \gamma V_o, \quad \gamma = \frac{R_2}{R_1 + R_2}.$$

The inverting input is set by the RC branch charging toward the output:

$$C \frac{dV_2}{dt} + \frac{V_2 - V_o}{R} = 0 \Rightarrow \frac{dV_2}{dt} + \frac{V_2}{RC} = \frac{V_o}{RC}.$$

The two saturation cases. When $V_2 < V_1$ the output holds at $V_o = +V_{sat}$ and V_2 charges toward $+V_{sat}$:

$$V_2(t) = V_{sat} + (V_2(0) - V_{sat})e^{-t/RC}.$$

When $V_2 > V_1$ the output holds at $V_o = -V_{sat}$ and V_2 decays toward $-V_{sat}$:

$$V_2(t) = -V_{sat} + (V_2(0) + V_{sat})e^{-t/RC}.$$

Each time V_2 reaches the reference $V_1 = \gamma V_o$ the comparator switches and the other case takes over, so the circuit oscillates.

Period and frequency. Taking one half cycle that starts at $V_2(0) = -\gamma V_{sat}$ with $V_o = +V_{sat}$, the switch happens when $V_2(t) = +\gamma V_{sat}$:

$$\gamma V_{sat} = V_{sat} + (-\gamma V_{sat} - V_{sat})e^{-t/RC} \Rightarrow e^{-t/RC} = \frac{1 - \gamma}{1 + \gamma}.$$

Solving for the half-period and using $\tau = RC$:

$$t = -\tau \ln\left(\frac{1 - \gamma}{1 + \gamma}\right), \quad T = 2t, \quad f = \frac{1}{T}.$$

The frequency depends only on τ and γ , so choosing R_1 and R_2 sets the blink rate.

Component selection. The design targets are $f \approx 1$ Hz with $RC = 1$ s and $R_1 + R_2 \approx R = 10$ k Ω . From $RC = 1$ s with $R = 10$ k Ω ,

$$C = \frac{1}{R} = \frac{1}{10 \text{ k}\Omega} = 100 \text{ }\mu\text{F}.$$

Setting $f = 1$ Hz (so $T = 1$ s, $t = 0.5$ s) with $\tau = 1$ s gives

$$\frac{1 - \gamma}{1 + \gamma} = e^{-0.5} = 0.6065 \Rightarrow \gamma = 0.245.$$

With $R_1 + R_2 = 10$ k Ω this is $R_2 \approx 2.45$ k Ω , $R_1 \approx 7.55$ k Ω . The nearest standard values are $R_1 = 7.5$ k Ω and $R_2 = 2.4$ k Ω , which give $\gamma = 0.2424$ and a predicted $f \approx 1.01$ Hz.

Expected node voltages. Using the ideal saturation level $V_{sat} = 5$ V: V_1 is a square wave that sits at $\pm \gamma V_{sat} = \pm 1.21$ V, and V_2 ramps exponentially between the same two limits. Both therefore have $V_{pp} = 2\gamma V_{sat} = 2.42$ V.

```

// Lab 7 - relaxation oscillator design check
R1 = 7.5e3;           // non-inverting divider, top
R2 = 2.4e3;           // non-inverting divider, bottom
R  = 10e3;            // RC feedback resistor
C  = 100e-6;          // RC feedback capacitor

gamma = R2 / (R1 + R2);
tau   = R * C;
thalf = -tau * log((1 - gamma) / (1 + gamma)); // one saturation case
T      = 2 * thalf;
f      = 1 / T;

mprintf("gamma = %.4f\n", gamma);
mprintf("tau   = %.4f s\n", tau);
mprintf("T      = %.4f s\n", T);
mprintf("f      = %.4f Hz\n", f);

```

Listing 1: SciLab design check for the oscillator frequency

```

gamma = 0.2424
tau   = 1.0000 s
T      = 0.9892 s
f      = 1.0109 Hz

```

Listing 2: SciLab runtime output

Task 2 targets. For 2 Hz ($t = 0.25$ s) at the original $\tau = 1$ s, the required ratio is

$$\frac{1 - \gamma}{1 + \gamma} = e^{-0.25} = 0.7788 \Rightarrow \gamma \approx 0.124.$$

After halving R to 5 k Ω the time constant drops to $\tau = 0.5$ s, so holding 2 Hz requires

$$\frac{1 - \gamma}{1 + \gamma} = e^{-0.5} = 0.6065 \Rightarrow \gamma \approx 0.245.$$

So halving the feedback resistor calls for a larger γ (a larger R_2 fraction) to keep the same blink rate; these two values are the targets for the measured potentiometer settings.

Simulations

The oscillator was simulated in LTSpice using the **UniversalOpAmp2** part, powered by two DC sources wired to ground with opposite polarity to create the $+V_{cc}$ and $-V_{cc}$ rails at 5 V and -5 V. The directive `.ic V(Vo)=1` seeds the output so the circuit starts switching, and a transient run of three to five periods (a few seconds) shows the steady oscillation of V_o , V_1 , and V_2 .

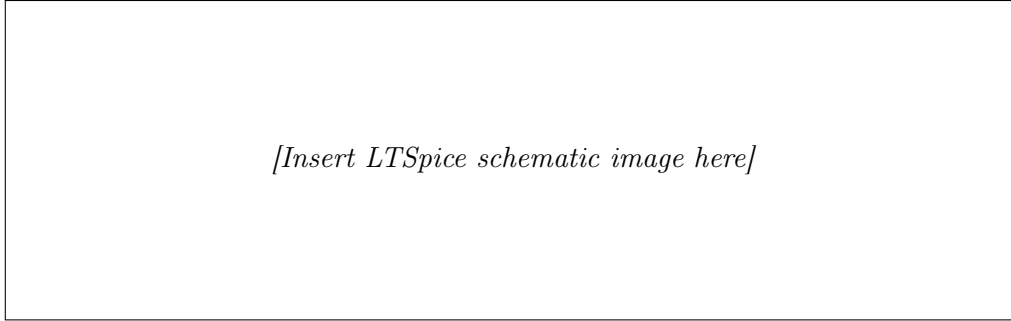


Figure 1: LTSpice schematic of the op-amp relaxation oscillator.

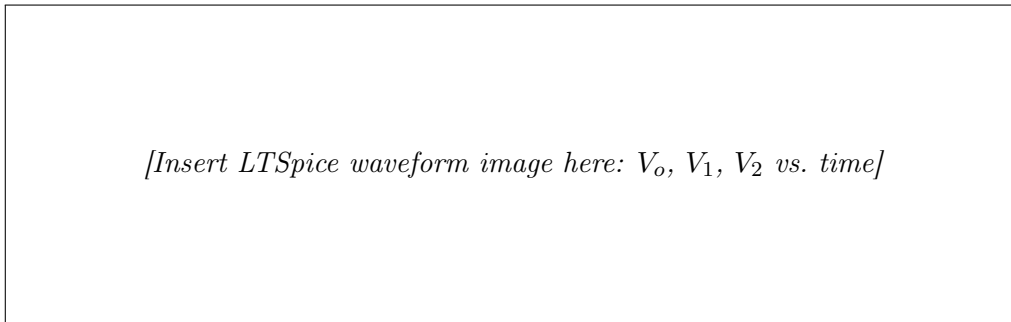


Figure 2: Simulated transient waveforms showing the output square wave V_o , the divider reference V_1 , and the exponentially charging node V_2 .

```
* Lab 7 - transient response oscillator
* Paste your exported LTSpice netlist here
.ic V(Vo)=1
.tran 0 5 0
.end
```

Listing 3: LTSpice netlist (paste exported netlist here)

Measurements

Task 1. Measured frequency and peak-to-peak voltage at the two input nodes, compared against the prelab predictions. The oscilloscope read V_1 and V_2 at essentially the same frequency (961.5 mHz and 961.1 mHz).

Table 1: Task 1 – measured node voltages vs. calculated ($V_{sat} = 5$ V).

Quantity	Calculated	Measured	% Error
Frequency	1.01 Hz	0.962 Hz	−4.9%
V_1 peak-to-peak	2.42 V	2.14 V	−11.6%
V_2 peak-to-peak	2.42 V	1.22 V	−49.6%

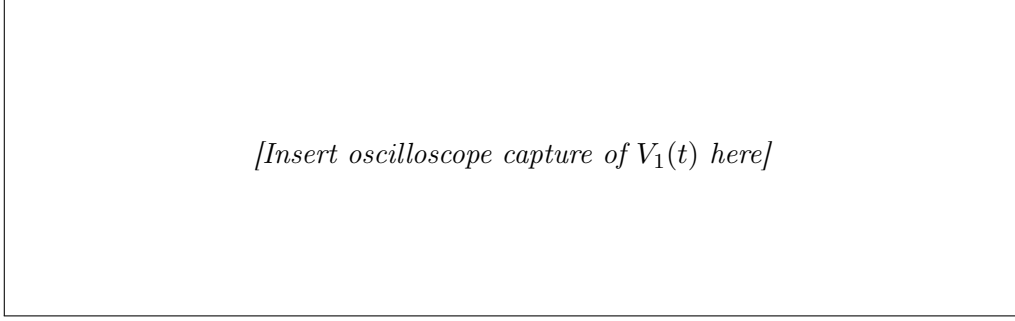


Figure 3: Measured $V_1(t)$ at the non-inverting input (Task 1).

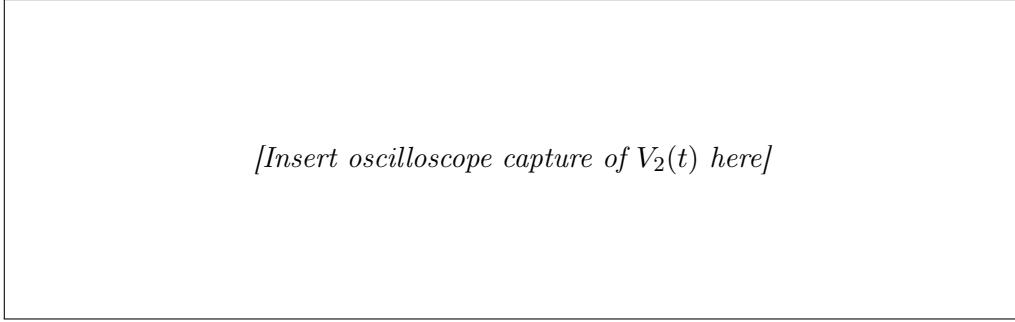


Figure 4: Measured $V_2(t)$ at the inverting input (Task 1).

Task 2. Potentiometer settings that produced 2 Hz oscillation, with the resulting γ . Here R_2 is the ground-side segment of the pot (the V_1 -to-ground resistor) and R_1 is the feedback segment (V_o to V_1), so that $\gamma = R_2/(R_1 + R_2)$ as defined in the prelab. For Step 2 the RC resistor was set to a standard 4.7 k Ω rather than an exact halving to 5 k Ω , so the 2 Hz target shifts from the prelab's $\gamma \approx 0.245$ to $\gamma \approx 0.260$.

Table 2: Task 2 – potentiometer values at 2 Hz.

Configuration	R_1	R_2	γ (measured)	γ (target)
Step 1: $R = 10 \text{ k}\Omega$	8.71 k Ω	1.34 k Ω	0.133	0.124
Step 2: $R = 4.7 \text{ k}\Omega$	7.72 k Ω	2.49 k Ω	0.244	0.260

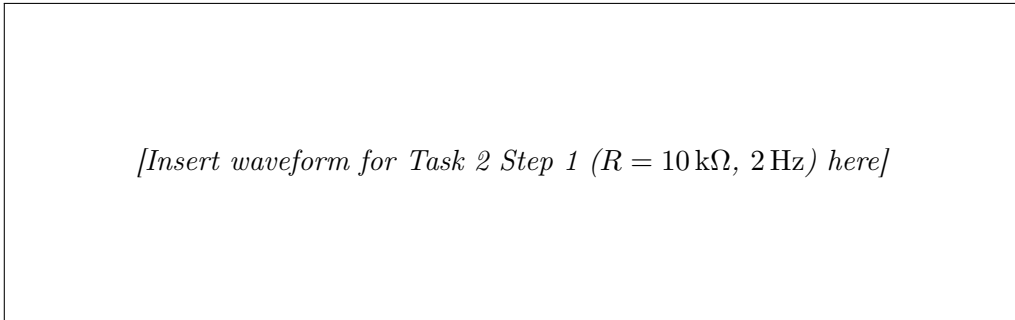


Figure 5: Task 2 Step 1 waveform at 2 Hz with the full feedback resistor.

[Insert waveform for Task 2 Step 2 ($R = 5\text{ k}\Omega$, 2 Hz) here]

Figure 6: Task 2 Step 2 waveform at 2 Hz with the feedback resistor halved.

Observations & Discussion

The circuit oscillated at 0.962 Hz, about 4.9 % below the predicted 1.01 Hz. Working backward from the measured period, the effective divider ratio was $\gamma \approx 0.254$ against the design value of 0.242, and the difference points to component tolerance: a slightly larger R_2 in the divider and a 100 μF electrolytic running above its nominal value both push $\tau = RC$ up and the frequency down. An electrolytic of this size commonly carries a tolerance near $\pm 20\%$, so a 4.9 % period shift is well within what the capacitor alone can account for.

The amplitudes tell a consistent story. The measured V_1 peak-to-peak was 2.14 V, about 11.6 % below the 2.42 V predicted with an ideal $V_{sat} = 5\text{ V}$. Since V_1 swings to $\pm\gamma V_{sat}$, the measured value implies an actual saturation level of about 4.4 V, meaning the 741 output stops roughly 0.6 V short of each rail rather than reaching them. This is the expected behavior of a real 741, whose output cannot swing fully to the supply rails, and it is the main reason the bench amplitudes sit below the simulated ones. The reduced swing also lowers the switching thresholds slightly, which feeds back into the small frequency offset noted above.

The one result that does not fit this picture is the V_2 peak-to-peak, which came out at 1.22 V, roughly half of V_1 . Because V_2 charges toward the output rail but reverses the moment it reaches the switching threshold $\pm\gamma V_{sat}$, which is the same level V_1 sits at, its peak-to-peak should be close to V_1 's 2.14 V rather than half of it. The most likely cause is a measurement issue on the V_2 channel, such as a vertical-scale or cursor error during the read, rather than a real circuit effect, and it would be worth re-capturing V_2 to confirm.

Task 2 returns to firmer ground and confirms the central prediction of the prelab. With the full 10 k Ω feedback resistor the pot settled at $\gamma = 0.133$ against a 2 Hz target of 0.124, and after dropping the RC resistor to 4.7 k Ω it settled at $\gamma = 0.244$ against a target of 0.260. Both land within about 7 % of target, consistent with the same resistor tolerance and capacitor offset seen in Task 1 together with the resolution of setting the wiper by hand. More telling than the individual numbers is the trend: lowering R shortens τ , which on its own would raise the frequency, so holding 2 Hz forces γ upward, and the measured values climb from 0.133 to 0.244 exactly as the theory requires.

Taken together, the dominant source of error is the electrolytic capacitor tolerance, which accounts for most of the frequency offset, followed by the non-ideal 741 saturation level, which explains the reduced amplitudes. Resistor tolerance, the finite input bias current of the 741 (so the zero-input-current assumption is only approximate), and, for Task 2, the precision of the potentiometer setting and the scope read-off all contribute at a smaller level. The one outlier is the V_2 peak-to-peak, which we attribute to a measurement error rather than any physical effect in the circuit.